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CSCI 4104

9/18/23
Problem Set 1

3.2-3 Is $2^{n+1} = O(2^n)$? Is $2^{2n} = O(2^n)$?

• Show if 2^{n+1} is $O(2^n)$, where $f(n) = 2^{n+1}$ and $g(n) = 2^n$:

Definition: $\exists c, k: 0 \leq f(n) \leq c \cdot g(n), \forall n > k$

So, show $\exists c, k: 0 \leq 2^{n+1} \leq c \cdot 2^n, \forall n > k$

Then, $2^{n+1} \leq c \cdot 2^n \rightarrow 2 \cdot 2^n \leq c \cdot 2^n \rightarrow 2 \leq c \cdot \frac{2^n}{2^n} \rightarrow 2 \leq c \rightarrow 2 \leq c$

We can choose $c=2$ and $k=1$ to satisfy the definition:

$0 \leq 2^{n+1} \leq 2 \cdot 2^n = 2^{n+1}, \forall n > 1$. Thus, $2^{n+1} = O(2^n)$ is true.

• Show if 2^{2n} is $O(2^n)$, where $f(n) = 2^{2n}$ and $g(n) = 2^n$:

Definition: $\exists c, k: 0 \leq f(n) \leq c \cdot g(n), \forall n > k$

So, show $\exists c, k: 0 \leq 2^{2n} \leq c \cdot 2^n, \forall n > k$

Then, $2^{2n} \leq c \cdot 2^n \rightarrow (2^n)^2 \leq c \cdot 2^n \rightarrow \frac{(2^n)^2}{2^n} \leq c \rightarrow 2^n \leq c$

The inequality $2^n \leq c$ is a contradiction, because the exponential growth function 2^n cannot be upper bounded by a constant "c". Thus, $2^{2n} = O(2^n)$ is false.

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16.1-3

i	cost
1	1
2	2
3	1
4	4
5	1
6	1
7	1
8	8
9	1
⋮	⋮

$$\sum_{i=0}^{\lfloor \lg(n) \rfloor} 2^i = 2^{\lfloor \lg(n) \rfloor + 1} - 1 \leftarrow \text{cost of powers of 2}$$

$$n - \lfloor \lg(n) \rfloor - 1 \leftarrow \text{cost of everything else}$$

→ Add terms and divide by "n" to find average

$$\text{cost: } \frac{2^{\lfloor \lg(n) \rfloor + 1} + n - \lfloor \lg(n) \rfloor - 1}{n} \leq \text{---}$$

$$\rightarrow \frac{2 \cdot 2^{\lg(n)} + n - \lg(n) - 2}{n} = \frac{2 \cdot n + n - \lg(n) - 2}{n}$$

$$= \frac{3n - \lg(n) - 2}{n} = O(1) \text{ is the amortized cost.}$$

3-2 $K \geq 1, \epsilon > 0, c > 1$ are constants

(e) $f(n) = n^{\lg c}, g(n) = c^{\lg n}$

$$n^{\lg c} = 2^{\lg(n^{\lg c})} = 2^{\lg(n) \cdot \lg(c)} = 2^{\lg(n) \cdot \lg(c)} = 2^{\lg(c^{\lg n})} = c^{\lg n}$$

Thus, $n^{\lg c} = c^{\lg n}$ which is an identity of logarithms

So, $\lim_{n \rightarrow \infty} \frac{n^{\lg c}}{c^{\lg n}} = \lim_{n \rightarrow \infty} 1 = 1$. $n^{\lg c} = O(c^{\lg n})$, $n^{\lg c} = \Omega(c^{\lg n})$, $n^{\lg c} = \Theta(c^{\lg n})$

(a) $f(n) = \lg^k n, g(n) = n^\epsilon$. Let $K=1$ and $\epsilon=2$, so that $f(n) = \lg n$ and $g(n) = n^2$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{\lg n}{n^2} = \lim_{n \rightarrow \infty} \frac{d/dn[\lg n]}{d/dn[n^2]} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n \ln 2}}{2n} = \lim_{n \rightarrow \infty} \frac{1}{\ln(2) \cdot 2 \cdot n^2} = \frac{1}{\infty} = 0$$

$$\lg^k n = o(n^\epsilon), \lg^k n = O(n^\epsilon), n^\epsilon = \omega(\lg^k n), n^\epsilon = \Omega(\lg^k n)$$

(b) $f(n) = n^K, g(n) = c^n$. Let $K=1$ and $c=2$, so that $f(n) = n$ and $g(n) = 2^n$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n}{2^n} = \lim_{n \rightarrow \infty} \frac{d/dn[n]}{d/dn[2^n]} = \lim_{n \rightarrow \infty} \frac{1}{\ln(2) \cdot 2^n} = \frac{1}{\infty} = 0$$

$$n^K = o(c^n), n^K = O(c^n), c^n = \omega(n^K), c^n = \Omega(n^K)$$

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(d) $f(n) = 2^n, g(n) = 2^{n/2}$
 $\lim_{n \rightarrow \infty} \frac{2^n}{2^{n/2}} = \lim_{n \rightarrow \infty} \frac{(2^n)^{1/2}}{(2^n)^{1/2}} = \lim_{n \rightarrow \infty} (2^n)^{1/2} = \lim_{n \rightarrow \infty} 2^{n/2} = 2^\infty = \infty$
 $2^{n/2} = O(2^n), 2^{n/2} = o(2^n), 2^n = \Omega(2^{n/2}), 2^n = \omega(2^{n/2})$

(e) $f(n) = n^{1.9}, g(n) = n^{1.5}$

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Algorithm Analysis

3-2 (HW problem)

(C)

Definition of
squeeze theorem

$$f(n) = \sqrt{n}, g(n) = n^{\sin(n)}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n^{\sin(n)}} = \lim_{n \rightarrow \infty} \frac{n^{1/2}}{n^{\sin(n)}} = \lim_{n \rightarrow \infty} n^{\frac{1}{2} - \sin(n)}$$

use squeeze theorem:

If $\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} g(n) = c$ and $f(n) \leq h(n) \leq g(n)$ for $\forall n$

Then, $\lim_{n \rightarrow \infty} h(n) = c$

$\sin(n)$
at lowest
is -1,
and highest
is 1

$n^{-1/2} \leq n^{1/2 - \sin(n)} \leq n^{3/2}$
Then, $\lim_{n \rightarrow \infty} n^{-1/2} = 0$ and $\lim_{n \rightarrow \infty} n^{3/2} = \infty$. So, $n^{\frac{1}{2} - \sin(n)}$ cannot be
bound asymptotically, because it doesn't follow the squeeze
theorem.

(F)

$$f(n) = \lg(n!), g(n) = \lg(n^n), n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\alpha_n}, \frac{1}{12n+1} < \alpha_n < \frac{1}{12n}$$

$$\lim_{n \rightarrow \infty} \frac{\lg(n!)}{\lg(n^n)} = \lim_{n \rightarrow \infty} \frac{\lg(n!)}{n \cdot \lg(n)} = \lim_{n \rightarrow \infty} \frac{\lg(\sqrt{2\pi n}) + \lg\left(\left(\frac{n}{e}\right)^n\right) + \lg(e^{\alpha_n})}{n \cdot \lg(n)}$$

$$= \lim_{n \rightarrow \infty} \frac{\lg(\sqrt{2\pi}) + \frac{1}{2} \lg(n) + n \cdot \lg\left(\frac{n}{e}\right) + \lg(e^{\alpha_n})}{n \cdot \lg(n)}$$

$$= \lim_{n \rightarrow \infty} \frac{\lg(\sqrt{2\pi}) + \frac{1}{2} \lg(n) + n \cdot \lg(n) - n \cdot \lg(e) + \alpha_n \lg(e)}{n \cdot \lg(n)}$$

$$= \lim_{n \rightarrow \infty} \frac{\lg(\sqrt{2\pi})}{n \cdot \lg(n)} + \frac{\frac{1}{2} \lg(n)}{n \cdot \lg(n)} + \frac{n \cdot \lg(n)}{n \cdot \lg(n)} - \frac{n \cdot \lg(e)}{n \cdot \lg(n)} + \frac{\alpha_n \lg(e)}{n \cdot \lg(n)}$$

$$= \frac{\lg(\sqrt{2\pi})}{\infty} + \frac{\frac{1}{2}}{\infty} + 1 - \frac{\lg(e)}{\infty} + \frac{0 \cdot \lg(e)}{\infty}$$

$$= 0 + 0 + 1 - 0 + 0 = 1$$

$$\lg(n!) = O(\lg(n^n)), \lg(n!) = \Omega(\lg(n^n)), \lg(n^n) = O(\lg(n!)),$$

$$\lg(n^n) = \Omega(\lg(n!)), \lg(n!) = \Theta(\lg(n^n)), \lg(n^n) = \Theta(\lg(n!))$$

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g) $f(n) = 3n+5, g(n) = 2n^2 - 4n + 1$
 $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{3n+5}{2n^2 - 4n + 1} = \lim_{n \rightarrow \infty} \frac{d/dn [3n+5]}{d/dn [2n^2 - 4n + 1]} = \lim_{n \rightarrow \infty} \frac{3}{4n-4} = \frac{3}{\infty} = 0$
 $3n+5 = o(2n^2 - 4n + 1), 3n+5 = O(2n^2 - 4n + 1), 2n^2 - 4n + 1 = \omega(3n+5),$
 $2n^2 - 4n + 1 = \Omega(3n+5)$

h) $f(n) = 4n-100, g(n) = 50 \lg(n)$
 $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{4n-100}{50 \lg(n)} = \lim_{n \rightarrow \infty} \frac{d/dn [4n-100]}{d/dn [50 \lg(n)]} = \lim_{n \rightarrow \infty} \frac{4}{50 \cdot \frac{1}{\ln(2) \cdot n}} = \lim_{n \rightarrow \infty} \frac{4}{50} \cdot \ln(2) \cdot n = \infty$
 $4n-100 = \omega(50 \lg(n)), 4n-100 = \Omega(50 \lg(n)), 50 \lg(n) = o(4n-100),$
 $50 \lg(n) = O(4n-100)$

i) $f(n) = 3 \lg n, g(n) = \lg(n^3)$
 By properties of logarithms $g(n) = \lg(n^3) = 3 \lg(n)$.
 So, $f(n) = 3 \lg n = g(n) \rightarrow f(n) = g(n)$
 $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{3 \lg n}{3 \lg n} = \lim_{n \rightarrow \infty} 1 = 1$
 $3 \lg n = O(\lg(n^3)), 3 \lg n = \Omega(\lg(n^3)), 3 \lg n = \Theta(\lg(n^3))$

j) $f(n) = n^3, g(n) = 100n^2$
 $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^3}{100n^2} = \lim_{n \rightarrow \infty} \frac{n}{100} = \infty$
 $n^3 = \omega(100n^2), n^3 = \Omega(100n^2), 100n^2 = o(n^3), 100n^2 = O(n^3)$

k) $f(n) = \lg n, g(n) = \sqrt{n}$
 $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{\lg n}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{d/dn [\lg n]}{d/dn [\sqrt{n}]} = \lim_{n \rightarrow \infty} \frac{\frac{1}{\ln(2) \cdot n}}{\frac{1}{2\sqrt{n}}} = \lim_{n \rightarrow \infty} \frac{2\sqrt{n}}{\ln(2) \cdot n} = \lim_{n \rightarrow \infty} \frac{2}{\ln(2) \cdot \sqrt{n}} = 0$
 $\lg n = o(\sqrt{n}), \lg n = O(\sqrt{n}), \sqrt{n} = \omega(\lg n), \sqrt{n} = \Omega(\lg n)$

L) $f(n) = 3n^3 + 2n^2 - 3n + 5, g(n) = 2n^3 - 3n^2 + 0.01n - 17$
 $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{3n^3 + 2n^2 - 3n + 5}{2n^3 - 3n^2 + 0.01n - 17} = \lim_{n \rightarrow \infty} \frac{\frac{3}{n^3} + \frac{2}{n^2} - \frac{3}{n} + \frac{5}{n^3}}{\frac{2}{n^3} - \frac{3}{n^2} + \frac{0.01}{n} - \frac{17}{n^3}} = \frac{3+0+0+0}{2-0+0-0} = \frac{3}{2}$

$3n^3 + 2n^2 - 3n + 5 = O(2n^3 - 3n^2 + 0.01n - 17),$
 $3n^3 + 2n^2 - 3n + 5 = \Omega(2n^3 - 3n^2 + 0.01n - 17),$
 $3n^3 + 2n^2 - 3n + 5 = \Theta(2n^3 - 3n^2 + 0.01n - 17)$

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Table of My Answers

	A	B	O	o	Ω	ω	Θ
a.	$lg^k n$	n^e	Yes	Yes	No	No	No
b.	n^k	c^n	Yes	Yes	No	No	No
c.	$\sqrt{n}$	$n^{sm(n)}$	No	No	No	No	No
d.	2^n	$2^{n/2}$	No	No	Yes	Yes	No
e.	$n^{lg c}$	$c^{lg n}$	Yes	No	Yes	No	Yes
f.	$lg(n!)$	$lg(n^n)$	Yes	No	Yes	No	Yes
g.	$3n+5$	$2n^2+4n+1$	Yes	Yes	No	No	No
h.	$4n-100$	$50lg(n)$	No	No	Yes	Yes	No
i.	$3lg(n)$	$lg(n^3)$	Yes	No	Yes	No	Yes
j.	n^3	$100n^2$	No	No	Yes	Yes	No
k.	$lg(n)$	$\sqrt{n}$	Yes	Yes	No	No	No
L.	$\uparrow$	$\uparrow$	Yes	No	Yes	No	Yes
$(3n^3+7n^2-3n+5)$		$(2n^3-3n^2+0.01n-17)$					